

Original Article

Retention of baricitinib-treated patients with rheumatoid arthritis in Japan: Interim results from a prospective observational cohort study

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Abstract

Objectives: To assess the retention rate, changes in disease activity and patient-reported outcomes (PROs), and predictors of treatment discontinuation among baricitinib-treated patients with rheumatoid arthritis (RA) in routine clinical practice in Japan.

Methods: This ongoing, multi-centre, prospective, observational study (starting February 2021) included ≥ 20 -year-old baricitinib-treated patients with RA. Baricitinib's retention rate and effectiveness were evaluated until 12 months. The association between patient characteristics and retention was evaluated using Random Forest analysis.

Results: This interim analysis included 353 patients (mean [standard deviation; SD] age 63.4 [13.6] years; 70.0% female). The 12-month retention rate was 70.2% (95% confidence interval [CI] 65.1, 74.6). The 12-month remission rates were 35.3%, 41.1%, and 79.6% per Clinical Disease Activity Index, Simplified Disease Activity Index, and Disease Activity Score with 28 joints–C-reactive protein. The mean (SD) changes in PROs from baseline scores were: physician global assessment of disease activity (-33.8 [27.8]), patient global assessment of disease activity (-18.3 [29.4]), Health Assessment Questionnaire-Disability Index (-0.33 [0.53]), and pain (-19.4 [27.0]). Ten variables influenced the model's prediction of treatment discontinuation.

Conclusions: Evidence from this interim analysis suggested the favourable effectiveness and drug survival of baricitinib in real-world Japanese patients with RA.

Keywords: baricitinib, CDAI, DAS28-CRP, HAQ-DI, retention, real world, rheumatoid arthritis, SDAI

Introduction

Rheumatoid arthritis (RA), a systemic autoimmune inflammatory disease that primarily affects the joints, is associated with progressive disability, premature death, and high socioeconomic costs.^{1,2} Treat-to-target (T2T) is the

standard-of-care (SOC) for RA, and the treatment goal is remission or its alternative, low disease activity (LDA), to prevent joint destruction and achieve low disability, leading to better health-related quality of life (HRQoL).^{3,4} Given the impact of RA on HRQoL, evaluating patients' perspectives on disease impact has been increasingly recognised.⁵ Hence, in addition to T2T, long-term patient satisfaction and treatment retention are critical measures of RA treatment success,⁶ making the evaluation of the long-term retention rate and patient-reported outcome (PRO) improvement, meaningful. However, despite the evolving landscape of clinical management for RA, including the availability of numerous pharmacological options (conventional synthetic disease-modifying antirheumatic drugs [csDMARDs], biological DMARDs [bDMARDs], and targeted synthetic DMARDs [tsDMARDs]), some patients with RA fail to reach or maintain the recommended treatment targets.^{2,3,4,7}

Baricitinib, a once-daily oral Janus kinase (JAK) inhibitor highly selective for JAK1 and JAK2, was approved in Japan in July 2017 for the treatment of RA in patients with inadequate response or intolerance to conventional RA treatments.⁸ In clinical trials and real-world studies, baricitinib has shown efficacy and effectiveness in disease activity control with rapid response and long-term sustainability, along with improvement in PROs.^{8,9,10,11,12}

Randomised controlled trials (RCTs) are the gold standard for assessing new therapies, but their rigorous eligibility criteria may limit external validity. This may lead to disparities between the efficacy observed in RCTs and real-world settings. After the approval of baricitinib, a post-marketing surveillance (PMS) study was implemented in Japan to evaluate baricitinib's effectiveness for up to 6 months,⁸ and persistence and safety for up to 3 years.¹³ These studies reported the incidence of deaths, adverse events, and adverse events of special interest such as herpes zoster, hepatic function disorder, serious infection, malignancy, major adverse cardiovascular events, and venous thromboembolism.^{8,13} The difference in the average age of patients prescribed baricitinib (63.9 years in the PMS study vs. 52.7 years in clinical trials) highlights the gaps in patient characteristics between RCTs and routine clinical settings.

As RA treatment generally continues for many years, drug retention is essential for optimal treatment effectiveness.^{6,14,15} Moreover, since the PMS study did not assess the long-term effectiveness of baricitinib,^{8,13} the long-term impact of baricitinib on disease activity and PROs in routine clinical practice in Japan, i.e., beyond 6 months, is unknown. Hence, it is important to compile real-world data regarding the on-label use of baricitinib in this setting. The current ongoing, prospective, observational study was conducted to assess the retention rate, changes in disease activity, and PROs among patients with RA treated with baricitinib, in routine clinical practice in Japan. Here, we provide data on

an interim analysis of the study to understand how the patients responded to baricitinib after 12 months of follow-up. We also examined the association between baseline characteristics and retention with baricitinib.

Methods

Study design

This multi-centre, prospective, observational cohort study started in February 2021. Patients were enrolled from March 2021 to December 2022 and observed till December 2024. Data were collected at baseline (defined as before the initiation of treatment with baricitinib) and at routine clinical care visits, i.e., post-baseline visits at approximately 1, 3, 6, 12, 18, and 24 months after the initiation of baricitinib treatment (**Figure 1**). Participating physicians enrolled patients based on their treatment decisions during routine clinical practice until the sample size of 340 was reached. Both patients and physicians provided informed consent. The decision to start treatment with baricitinib was made according to the routine SOC before enrolment in the study, with no attempt to influence the prescribing patterns of any individual physician. Treatment regimens and any modifications to them, including discontinuation of baricitinib, were solely at the discretion of the physician and the patient. Participation in the study did not affect the co-payment for any treatment received by patients during the study.

Study population

This study included Japanese patients aged ≥ 20 years who were diagnosed with RA as defined by the American College of Rheumatology/ European League Against Rheumatism 2010 Criteria for the Classification of RA¹⁶ and initiated baricitinib during routine clinical care for the treatment of RA from March 2021 to December 2022. The exclusion criteria were: a) received previous treatment with baricitinib; b) enrolled in other clinical trials during this study; c) enrolled in the baricitinib PMS study; d) diagnosed with other diseases, such as but not limited to, psoriatic arthritis, fibromyalgia, Sjogren syndrome, and any other systemic inflammatory condition “which make it difficult to appropriately assess RA activity in a physician’s opinion”; e) could not understand how to answer the relevant patient questionnaires, give written consent in Japanese, and follow study-related instructions; f) employees of Eli Lilly and Company or any of its subsidiaries; and g) study-affiliated site personnel and/or their immediate families (defined as a spouse, parent, child, or sibling [whether biological or through legal adoption]).

Some patients participated in a Customer Support Programme (CSP) from the T2 visit at 1 month. CSP is a no-cost service that offers support throughout the treatment journey by connecting patients with a dedicated “companion in care” representative who can answer questions and address concerns. Patients can contact the representative via social networking service applications, or a phone number provided by Eli Lilly Japan K.K.

Data collection and outcomes

Data were collected using internationally validated tools.¹⁷⁻²¹ Participating physicians transcribed the data from the patients’ medical records into an electronic data capture system directly at the physicians’ site. Questionnaires were distributed to the recruited patients who consented to participate at each institution. The baricitinib dose, the use of any concomitant treatment for RA (including biological/ targeted synthetic disease-modifying antirheumatic drugs [b/tsDMARDs] not recommended for concurrent use with baricitinib), and any changes and/or additions to therapy during the follow-up period, were recorded.

Patients who discontinued baricitinib or were lost to follow-up were discontinued from the study. The reason(s) for discontinuation of RA treatment with baricitinib were recorded by category - primary non-response, secondary loss of response, adverse event, sustained clinical response, non-compliance, patient decision, physician decision, cannot afford medication, or “other”. This study did not collect details about adverse events after treatment initiation. Treatment termination due to sustained clinical response was not considered an event of discontinuation but was treated as censored in the time to discontinuation (TTD) analysis.

Demographic and clinical information, such as RA treatment history at baseline, current RA treatment from baseline, and CSP participation status from the T2 visit at 1 month, were collected. The retention rate of baricitinib was assessed till 12 months of follow-up in this interim analysis. Disease activity (Clinical Disease Activity Index [CDAI],¹⁷ Simplified Disease Activity Index [SDAI],¹⁷ and Disease Activity Score modified to include the 28 diarthrodial joint count – C-reactive protein [DAS28-CRP]),^{18,19} and health outcomes, including the Health Assessment Questionnaire-Disability Index (HAQ-DI),^{20,21} Physician’s/Patient’s Global Assessment of Disease Activity VAS, and pain related to RA as measured by the patient’s assessment of pain (0–100 mm visual analogue scale [VAS]) (**Supplementary Table 1**) were assessed at 1, 3, 6, and 12 months. Disease activity was categorised as remission, LDA, moderate disease activity (MDA), or high disease activity (HDA) using standard cut-off values.²²

Statistical analysis

The analysis included all eligible patients who provided consent and for whom the physician signed the electronic case report form (CRF). Patients were excluded from the analysis if: a) the electronic CRF was not collected, b) they were mistakenly registered, c) they were re-registered after the study was stopped, d) baricitinib administration could not be confirmed, e) all efficacy-related pages in the electronic CRF were not signed, f) they did not have a history of RA treatment, and g) any deviations from the enrolment criteria were observed. Data up to the last data collection point were included for patients who were lost to follow-up or dropped out of the study. Since this is a 12-month interim analysis, patients who continued baricitinib at 12 months were censored.

Descriptive summary statistics including mean (standard deviation [SD]) for continuous variables and proportions for categorical variables, were reported. Changes from baseline were presented for outcomes at each relevant visit. Analysis was based on the observed values of patients at each time point. As secondary analyses, modified baseline observation carried forward (mBOCF) for continuous variables and non-responder imputation (NRI) for categorical variables were also conducted. The mBOCF imputation was applied to data missing at the discontinuation visit or time points after discontinuation. All patients who discontinued the study because of primary non-response, secondary loss of response, adverse events, patient decision and/or physician decision, were defined as non-responders at the time of discontinuation or subsequent time points for evaluation. The baseline observation was carried forward to subsequent time points for evaluation of patients who discontinued the study treatment because of primary non-response, secondary loss of response, adverse events, patient decision and/or physician decision, and/or discontinuations for reasons unrelated to treatment (e.g. cannot afford medication, non-compliance or other). The observed data were used for subsequent time points for evaluation of patients who discontinued the study treatment because of sustained clinical response.

TTD was presented as a Kaplan-Meier curve. TTD was also evaluated in subgroups based on prior b/tsDMARD use and CSP participation status at T2–T4. Patients who discontinued treatment due to sustained clinical response were censored. In addition, machine learning Random Forest was used to assess the association between retention and patient characteristics. Random Forest is a popular classification algorithm in statistical analysis.^{23,24} It involves constructing multiple decision trees with random sampling and combining the outputs from the trees. The use of randomness enhances the predictive strength of the model. In this study's model, discontinuation was the response variable and the patient characteristics were baseline pain (VAS), HAQ-DI, and CDAI scores (all continuous); duration of RA from diagnosis

in years (<5 years or ≥5 years); prior b/tsDMARDs (naïve, 1, 2, or ≥2); sex (male or female); RA support app usage status (yes or no); nicotine use (yes or no); oral glucocorticoid use at baseline (yes or no); initial dose (2 mg or 4 mg); concomitant csDMARD use (yes or no); body mass index (BMI) in kg/m² (<18.5, ≥18.5 to <25, and ≥25); and age in years (<65 years or ≥65 years). These characteristics were represented based on its variable importance i.e. how much it contributed to the model.

Results

Patient demographic and clinical characteristics

As of March 15, 2024, among the 368 patients enrolled in the study, 353 were included in this interim analysis (**Figure 2**). Fifteen patients were excluded because their electronic CRF was not collected, informed consent was not obtained, or they did not meet the patient selection criteria. In the effectiveness analysis set (N=353), the mean (SD) age and BMI was 63.4 (13.6) and 23.1 (3.9); 79.0% were female (**Table 1**). The mean (SD) CDAI at baseline was 20.0 (12.1), and 41.9% and 34.3% had MDA and HDA, respectively. Some patients (23.8%) participated in the CSP at least once at T2–T4.

The median (Q1, Q3) duration of RA from diagnosis was 4.25 (1.22, 11.72) years (**Table 2**). A total of 30.9% of patients were naïve to b/tsDMARDs, while 65.7% and 13.0% had RA treatment history with any bDMARD or tsDMARD, respectively.

Retention rate

Overall, 88.4% (95% confidence interval [CI] 84.5, 91.3); 80.4% (95% CI 75.9, 84.2); and 70.2% (95% CI 65.1, 74.6) of patients remained persistent on baricitinib at 3, 6, and 12 months, respectively. A total of 106 patients (30.0%) discontinued baricitinib before 12 months. The most common reasons for treatment discontinuation were: i) at 0–90 days: primary non-response (5.7%), ii) at 91–180 days: others i.e. any reason except primary non-response, secondary non-response, and adverse events (2.3%), iii) at 181–270 days: secondary non-response (3.5%), and iv) at 271–365 days: secondary non-response (2.7%) (**Figure 3A**). A few patients (2.8%) discontinued treatment due to adverse events throughout the study period. The 12-month retention rates in b/tsDMARD-naïve (74.1% [95% CI 64.8, 81.3]) and b/tsDMARD-any (68.4% [95% CI 62.2, 73.8]) subgroups were similar (**Figure 3C**). The retention rates of patients who participated in the CSP at all visits was

85.7% (95% CI 62.0, 95.2), in those who participated at least once was 67.9% (95% CI 56.7, 76.7), and in those who participated at least twice was 68.0% (95% CI 53.2, 79.0) (**Supplementary Figure 1**).

Concomitant RA medication use

Among patients who used oral glucocorticoids concomitantly at baseline and continued baricitinib up to 12 months, 37.8% could discontinue the concomitant oral glucocorticoid at 12 months (**Table 3**). On the other hand, among patients who used methotrexate concomitantly at baseline, 91.5% still received it at 12 months. Supplementary Table 2 presents the results of concomitant use of other csDMARDs.

Clinical outcomes (as observed analyses)

CDAI, SDAI, and DAS28-CRP

The mean (SD) CDAI score decreased from 20.0 (12.1) at baseline to 6.15 (6.15) at 12 months (**Table 4**). The CDAI score rapidly improved in a month after baricitinib treatment and almost plateaued after 3 months. The LDA and remission rates were 44.4% and 15.6% at 1 month and 49.2% and 24.2% at 3 months, respectively (**Table 5**). Similar rapid reductions from baseline to 12 months were also observed in mean (SD) SDAI (21.1 [12.7] to 6.27 [6.26]) and DAS28-CRP scores (3.73 [1.33] to 1.99 [0.90]) (**Table 4**). The LDA and remission rates for SDAI were 42.3% and 19.8% at 1 month and 48.8% and 27.1% at 3 months, respectively (**Table 5**). The LDA and remission rates for DAS28-CRP were 20.0% and 58.0% at 1 month and 11.0% and 71.5% at 3 months, respectively. In the mBOCF analysis, the mean (SD) scores decreased from baseline to 12 months for CDAI (20.0 [12.1] to 11.0 [11.5]), SDAI (21.1 [12.7] to 11.7 [12.2]), and DAS28-CRP (3.7 [1.3] to 2.6 [1.4]) (**Supplementary Table 3**). These changes were comparable to the observed analysis results.

Physician global assessment of disease activity

Mean (SD) physician global assessment of disease activity score decreased from 49.2 (24.2) at baseline to 14.3 (15.8) at 12 months. The score rapidly improved by 1 month (mean [SD] change -25.0 [27.7]) (**Table 6**). The mBOCF analysis showed a similar reduction in mean (SD) score from 49.2 (24.2) at baseline to 25.4 (26.2) at 12 months (**Supplementary Table 4**).

Patient-reported outcomes

Mean (SD) scores decreased from baseline to 12 months in HAQ-DI (0.96 [0.76] to 0.54 [0.65]), pain VAS (48.4 [26.5] to 25.2 [21.4]), and patient global assessment of disease activity (48.8 [25.2] to 27.7 [22.3]) (**Table 7**). Mean (SD) scores decreased from baseline to 12 months in the mBOCF analysis too for HAQ-DI (0.96 [0.76] to 0.75 [0.77]), pain VAS (48.4 [26.5] to 34.9 [27.6]), and patient global assessment of disease activity (48.4 [25.2] to 36.7 [26.9]) (**Supplementary Table 5**).

Association between patient characteristics and retention

We conducted a Random Forest analysis to evaluate the association between patient characteristics and retention. Among the 13 variables selected with existing medical knowledge, 10 variables led to better model prediction in terms of influence on treatment discontinuation: baseline pain score (VAS), baseline HAQ-DI score, prior b/tsDMARDs, duration of RA from diagnosis (years), RA support app usage status, sex, nicotine use, baseline CDAI score, oral glucocorticoids at baseline, and concomitant csDMARD (**Figure 4**).

The partial dependency plots (**Supplementary Figure 2**) for the top 2 influential variables for model prediction showed that the expected number of discontinuations increased along with an increase in baseline pain score (VAS) and HAQ-DI score.

Discussion

This interim analysis of a prospective observational study evaluated the retention rate of baricitinib for up to 12 months in routine clinical settings in Japan. The 12-month retention rate was 70%; primary non-response was the main reason for discontinuation. Old age is a risk factor for adverse events in baricitinib-treated patients.^{25,26} However, although ≥50.0% of the total population was ≥65 years old, only 2.8% of participants discontinued due to adverse events. In addition, baricitinib-treated patients showed early improvements in disease activity and PRO measures that were sustained for up to 12 months.

The current study's 12-month retention rate (70%) is in line with previous real-world studies that evaluated the persistence rate of baricitinib at timepoints ranging from 6 months to 3 years.^{8,11,13,27-29} At 6 months, the persistence rates from Japanese studies were 75.4% (PMS study)⁸ and 86.5% (a registry-based study),¹¹ while the multinational, prospective, observational RA-BE-REAL study reported the highest persistence rate (94.9%).²⁷ The 12-month persistence rate in the 3-year baricitinib PMS study (66.1%) was similar to our study's result.¹³ Insufficient effectiveness was the most common reason for discontinuation in the current (1.4%–5.7%) and other real-world studies of baricitinib (3.3%–21.3%).^{8,11,13,27,29}

In the current analysis, higher baseline pain (VAS) and HAQ-DI scores were associated with higher discontinuation based on the variable importance/partial plot of the Random Forest analysis. Other real-world studies from Italy have reported a greater number of previous b/tsDMARD therapies and high corticosteroid dosage to be associated with greater baricitinib discontinuation.^{30,31} These variables were also present in the current analysis' model.

The current study had more b/tsDMARD-naïve patients than the PMS study⁸ (30.9% vs. 10.2%). However, the clinical outcomes (CDAI, SDAI, and DAS28-CRP) were mostly similar. The proportion of patients achieving remission per CDAI and SDAI at 6 months was similar in the current study (29.4% and 32.0%) and the PMS study (31.5% and 33.0%).⁸ However, more patients achieved remission per DAS28-CRP at 6 months in the current study (73.4%) than in the PMS study (61.3%)⁸ and in a Japanese multicentre registry study (58.9%).¹¹ In the current study, substantial improvements in the PROs (pain VAS and HAQ-DI) occurred rapidly in a month and were sustained until month 12. Pain exceeded the tolerable level (VAS 40 mm)³² in >50% of the patients at baseline. However, the change in pain VAS score from baseline to 1 month (-15.7) exceeded the minimum clinically important improvement threshold in RA (-15),³² thus highlighting the rapid clinically meaningful improvement with baricitinib. Torikai et al³³ also reported similar rapid improvement in pain VAS and HAQ-DI in a Japanese real-world study of 32 patients with an inadequate response to DMARDs.

The Japan College of Rheumatology clinical practice guidelines for the management of RA recommends glucocorticoids as auxiliary treatment, especially for RA flares, but advises using them at the lowest dose and tapering off within a few months^{34,35} since glucocorticoids increase the risk of mortality, osteoporotic fractures, serious infections, diabetes, and other side effects.^{36,37} Real-world Italian studies have reported that JAK inhibitors such as baricitinib have a 'steroid-sparing effect', which helps to reduce glucocorticoid dose and speed up withdrawal, consequently decreasing the risk of glucocorticoid-related side effects.^{38,39} In the current study, 37.8% of patients using baricitinib and concomitant oral glucocorticoids at baseline could discontinue the oral glucocorticoid at 12 months. The RA-BEAM trial¹² showed

baricitinib's clinical superiority over adalimumab i.e., the SOC which also has a steroid-sparing effect.⁴⁰ Hence, the steroid-sparing effect of baricitinib in the real-world setting may be due to its high effectiveness in pain and disease activity.

Strengths and limitations

This study represents a comprehensive interim prospective report of the real-world effectiveness of baricitinib among adult patients in Japan with RA. Its major strength is the evaluation of retention, multiple clinical outcomes, and PROs over a year in a large number of patients. However, as this is an interim analysis of an ongoing study, the results should be interpreted with caution. This study also has a few limitations. The main analysis was based on the observed values, and the secondary analysis was based on mBOCF and NRI. The imputation may be biased since no clear mechanism of missingness of the clinical outcomes can be shown. Also, as this is an observational study, there were no control or comparator groups. Finally, the study was not designed to capture safety-related outcomes, so such safety data were not reported.

Conclusion

This interim analysis of an ongoing prospective observational study suggests the favourable effectiveness of baricitinib in real-world settings among the Japanese population with RA.

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Authorship/Author Contribution

All named authors meet the International Committee of Medical Journal Editors (ICMJE) criteria for authorship for this article, take responsibility for the integrity of the work as a whole, and have given their approval for this version to be published.

Data Availability Statement

The data in this article will be shared on reasonable request to the corresponding author. Part of this manuscript was accepted for presentation at the 69th Annual General Assembly and Scientific Meeting of the Japan College of Rheumatology.

Conflict of interests

YK received a grant or contract and honoraria from Eli Lilly Japan K.K. CW, NT, and MT are employees of Eli Lilly Japan K.K. and are minor stockholders of Eli Lilly and Company. KI received a grant or contract from Mitsubishi-Tanabe; consulting fees from Abbvie, Eli Lilly Japan K.K., Eisai, Gilead, and Asahi Kasei; speaker's fees from Eli Lilly Japan K.K., Abbvie, Eisai, Gilead, AstraZeneca, Asahi Kasei, Pfizer, Chugai, Mitsubishi-Tanabe, BMS, Janssen, and Novartis.

Compliance with Ethics Guidelines

All study data were anonymised and informed consent was obtained from the patients. The study was conducted in accordance with ethical principles that have their origin in the Declaration of Helsinki and are consistent with Good Pharmacoeconomics Practices, as well as applicable laws and regulation of Japan.

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Tables

Table 1. Baseline demographic and clinical characteristics (N=353)

Variable		
Age (years)	n	353
	Mean (SD)	63.4 (13.6)
Sex, n (%)	Female	279 (79.0)
Height (cm)	n	337
	Mean (SD)	156.7 (7.9)
Weight (kg)	n	341
	Mean (SD)	57.2 (11.8)
BMI (kg/m²)	n	335
	Mean (SD)	23.1 (3.9)
CSP participation status at T2-T4, n (%)		
	all Yes	21 (5.9)
	at least once Yes	84 (23.8)
	at least twice Yes	50 (14.2)
CDAI	n	335
	Mean (SD)	20.0 (12.1)
CDAI by categories, n (%)		
	Remission (CDAI ≤2.8)	9 (2.5)
	LDA (2.8< CDAI ≤10)	57 (16.1)
	MDA (10< CDAI ≤22)	148 (41.9)
	HDA (22< CDAI)	121 (34.3)
	Missing	18 (5.1)

BMI, body mass index; CDAI, Clinical Disease Activity Index; CSP, Customer Support Programme, HDA, high disease activity; LDA, low disease activity; MDA, moderate disease activity; N, total population size; n, sample size; SD, standard deviation

Table 2. RA history and RA treatment (N=353)

Variable		
Duration of RA from diagnosis (years)		
	Median (Q1, Q3)	4.25 (1.22, 11.72)
Duration of RA from diagnosis (years) by categories, n (%)		
	<1	78 (22.1)
	1 ≤ <5	116 (32.9)
	5 ≤ <10	54 (15.3)
	10 ≤ <20	72 (20.4)
	20 ≤	33 (9.3)
RA treatment, n (%)		
RA treatment with b/tsDMARDs at any time before enrollment		
	Naïve to b/tsDMARDs	109 (30.9)
	Any bDMARD	232 (65.7)
	Any tsDMARD	46 (13.0)
Number of previous b/tsDMARDs received at any time before enrollment		
	Naïve to b/tsDMARDs	109 (30.9)
	One b/tsDMARD	128 (36.3)
	Two b/tsDMARDs	66 (18.7)
	More than two b/tsDMARDs	50 (14.2)
Concomitant RA medications (other than glucocorticoids)		
	Methotrexate	171 (48.4)
	Salazosulfapyridine	46 (13.0)
	Iguratimod	48 (13.6)
	Tacrolimus	8 (2.3)
	Leflunomide	3 (0.8)

Concomitant oral glucocorticoids at baseline		
Yes		116 (32.9)
No		236 (66.9)
Missing		1 (0.3)

b/tsDMARD, biological/ targeted synthetic disease-modifying antirheumatic drugs; N, total population size; n, sample size; RA, rheumatoid arthritis

Table 3. Concomitant use of oral glucocorticoids and methotrexate at baseline and follow-up visits

Timepoint	Number of patients at each timepoint (N=353)	Number of patients using the medication at each timepoint, n (%)
Oral glucocorticoids		
Baseline	116	116 (100.0)
1 month	100	98 (98.0)
3 months	96	87 (90.6)
6 months	84	66 (78.6)
12 months	74	46 (62.2)
Methotrexate		
Baseline	165	165 (100.0)
1 month	139	136 (97.8)
3 months	143	137 (95.8)
6 months	128	123 (96.1)
12 months	117	107 (91.5)

N, total population size; n, sample size

Table 4. Disease activity scores and score changes from baseline over time (as observed analysis)

		CDAI (N=353)		SDAI (N=353)		DAS28-CRP (N=353)	
		Value	Change from baseline	Value	Change from baseline	Value	Change from baseline
Baseline	n	335	-	331	-	331	-
	Mean (SD)	20.03 (12.08)	-	21.13 (12.74)	-	3.73 (1.33)	-
	Median	18.00	-	18.60	-	3.57	-
	[Q1, Q3]	[11.50, 26.30]	-	[11.70, 27.80]	-	[2.78, 4.69]	-
	[Min, Max]	[0.5, 72.9]	-	[0.6, 77.2]	-	[1.1, 7.7]	-
1 month	n	270	259	248	234	250	236
	Mean (SD)	10.40 (9.39)	-9.77 (12.14)	10.94 (10.00)	-10.30 (12.64)	2.58 (1.17)	-1.20 (1.32)
	Median	8.00	-7.60	8.30	-8.40	2.36	-1.06
	[Q1, Q3]	[4.00, 13.60]	[-15.80, -1.80]	[4.10, 14.25]	[-17.60, -1.07]	[1.67, 3.13]	[-2.17, -0.21]
	[Min, Max]	[0.0, 65.0]	[-62.7, 45.0]	[0.0, 66.1]	[-63.7, 36.9]	[1.0, 7.1]	[-4.7, 2.3]
3 months	n	256	242	240	223	246	228
	Mean (SD)	8.22 (8.36)	-11.68 (13.11)	8.69 (8.64)	-12.14 (14.05)	2.32 (1.09)	-1.40 (1.47)
	Median	6.15	-10.05	6.40	-10.10	2.05	-1.40
	[Q1, Q3]	[3.00, 10.75]	[-17.50, -3.00]	[3.10, 10.85]	[-19.60, -2.80]	[1.50, 2.76]	[-2.45, -0.32]
	[Min, Max]	[0.0, 61.3]	[-71.0, 42.3]	[0.0, 62.1]	[-72.0, 43.1]	[1.0, 6.7]	[-6.2, 3.8]

6 months	n	231	220	219	205	222	208
	Mean (SD)	7.64 (7.67)	-11.75 (13.12)	7.82 (7.87)	-12.47 (14.05)	2.23 (1.00)	-1.42 (1.44)
	Median	5.80	-10.05	5.50	-10.30	2.01	-1.33
	[Q1, Q3]	[2.10, 10.00]	[-17.95, -2.45]	[2.30, 10.20]	[-20.30, -2.30]	[1.48, 2.71]	[-2.33, -0.25]
	[Min, Max]	[0.0, 48.0]	[-71.7, 24.0]	[0.0, 48.0]	[-72.6, 24.0]	[1.0, 5.8]	[-6.1, 2.1]
12 months	n	204	194	197	184	201	187
	Mean (SD)	6.15 (6.15)	-13.12 (12.26)	6.27 (6.26)	-13.85 (12.63)	1.99 (0.90)	-1.67 (1.37)
	Median	4.00	-10.40	4.00	-10.80	1.75	-1.59
	[Q1, Q3]	[2.00, 8.70]	[-19.10, -5.00]	[2.00, 8.50]	[-21.20, -5.15]	[1.34, 2.42]	[-2.59, -0.68]
	[Min, Max]	[0.0, 36.4]	[-69.0, 22.5]	[0.0, 37.8]	[-70.0, 22.4]	[1.0, 5.2]	[-6.0, 2.5]

CDAI, Clinical Disease Activity Index; DAS28-CRP, Disease Activity Score modified to include the 28 diarthrodial joint count - C-reactive protein; N, total population size; n, sample size; SDAI, Simplified Disease Activity Index; SD, standard deviation

Table 5. Patients achieving low disease activity or remission (as observed analysis)

		CDAI (N=353) (n [%])	SDAI (N=353) (n [%])	DAS28-CRP (N=353) (n [%])
1 month	n	270	248	250
	HDA	25 (9.3)	18 (7.3)	11 (4.4)
	MDA	83 (30.7)	76 (30.6)	44 (17.6)
	LDA	120 (44.4)	105 (42.3)	50 (20.0)
	Remission	42 (15.6)	49 (19.8)	145 (58.0)
3 months	n	256	240	246
	HDA	14 (5.5)	11 (4.6)	8 (3.3)
	MDA	54 (21.1)	47 (19.6)	35 (14.2)
	LDA	126 (49.2)	117 (48.8)	27 (11.0)
	Remission	62 (24.2)	65 (27.1)	176 (71.5)
6 months	n	231	219	222
	HDA	12 (5.2)	8 (3.7)	4 (1.8)
	MDA	44 (19.0)	43 (19.6)	35 (15.8)
	LDA	107 (46.3)	98 (44.7)	20 (9.0)
	Remission	68 (29.4)	70 (32.0)	163 (73.4)
12 months	n	204	197	201
	HDA	8 (3.9)	3 (1.5)	1 (0.5)
	MDA	33 (16.2)	30 (15.2)	21 (10.4)
	LDA	91 (44.6)	83 (42.1)	19 (9.5)
	Remission	72 (35.3)	81 (41.1)	160 (79.6)

For CDAI, SDAI, and DAS28-CRP, respectively: high disease activity are scores >22, >26, and >5.1; middle disease activity are scores >10–≤22, >11–≤26, and ≥3.2–≤5.1; low disease activity are scores >2.8–≤10, >3.3–≤11, and ≥2.6–<3.2; and remission are scores ≤2.8, ≤3.3, and <2.6.

CDAI, Clinical Disease Activity Index; DAS28-CRP, Disease Activity Score modified to include the 28 diarthrodial joint count - C-reactive protein; HDA, high disease activity; LDA, low disease activity; MDA, moderate disease activity; N, total population size; n, sample size; SDAI, Simplified Disease Activity Index.

Table 6. Physician Global assessment of disease activity (VAS) scores and score changes from baseline over time (as observed analysis)

		Physician Global assessment of disease activity	
		Value	Change from baseline
Baseline	n	343	-
	Mean (SD)	49.2 (24.2)	-
	Median	50.0	-
	[Q1, Q3]	[30.0, 69.0]	-
	[Min, Max]	[0, 100]	-
1 month	n	280	273
	Mean (SD)	23.1 (20.9)	-25.0 (27.7)
	Median	18.0	-23.0
	[Q1, Q3]	[8.0, 31.5]	[-41.0, -7.0]
	[Min, Max]	[0, 92]	[-97, 72]
3 months	n	281	274
	Mean (SD)	17.1 (18.1)	-31.1 (28.6)
	Median	11.0	-30.0
	[Q1, Q3]	[5.0, 23.0]	[-50.0, -10.0]
	[Min, Max]	[0, 100]	[-100, 70]
6 months	n	249	243
	Mean (SD)	17.4 (18.3)	-31.4 (29.0)
	Median	10.0	-30.0
	[Q1, Q3]	[5.0, 24.0]	[-51.0, -10.0]

	[Min, Max]	[0, 89]	[-100, 45]
12 months	n	227	222
	Mean (SD)	14.3 (15.8)	-33.8 (27.8)
	Median	10.0	-31.0
	[Q1, Q3]	[1.0, 20.0]	[-52.0, -14.0]
	[Min, Max]	[0, 74]	[-100, 50]

n, sample size; SD, standard deviation; VAS, visual analogue scale

Table 7. Patient-reported outcomes scores and score changes from baseline over time (as observed analysis)

		HAQ-DI (N=353)		Pain VAS (N=353)		Patient global assessment of disease activity (N=353)	
		Value	Change from baseline	Value	Change from baseline	Value	Change from baseline
Baseline	n	320	-	332	-	342	-
	Mean (SD)	0.96 (0.76)	-	48.4 (26.5)	-	48.8 (25.2)	-
	Median	0.88	-	50.0	-	50.0	-
	[Q1, Q3]	[0.38, 1.50]	-	[28.5, 70.0]	-	[30.0, 70.0]	-
	[Min, Max]	[0.00, 3.00]	-	[0, 100]	-	[0, 100]	-
1 month	n	265	244	266	256	283	277
	Mean (SD)	0.78 (0.74)	-0.19 (0.46)	32.9 (24.1)	-15.7 (25.9)	35.6 (25.2)	-13.6 (27.7)
	Median	0.63	-0.13	29.5	-12.0	30.0	-15.0
	[Q1, Q3]	[0.25, 1.13]	[-0.50, 0.00]	[11.0, 50.0]	[-31.5, 0.0]	[13.0, 50.0]	[-30.0, 1.0]
	[Min, Max]	[0.00, 3.00]	[-2.25, 1.00]	[0, 100]	[-85, 59]	[0, 100]	[-87, 62]
3 months	n	270	247	250	239	275	265
	Mean (SD)	0.65 (0.74)	-0.31 (0.57)	28.3 (24.2)	-21.5 (28.1)	30.9 (24.1)	-17.5 (29.7)
	Median	0.38	-0.25	20.0	-20.0	22.0	-15.0
	[Q1, Q3]	[0.13, 1.00]	[-0.63, 0.00]	[10.0, 43.0]	[-40.0, -1.0]	[10.0, 50.0]	[-39.0, 0.0]
	[Min, Max]	[0.00, 3.00]	[-2.13, 2.63]	[0, 100]	[-90, 50]	[0, 100]	[-90, 64]
6 months	n	238	217	223	211	242	235
	Mean (SD)	0.62 (0.70)	-0.31 (0.57)	27.2 (22.5)	-19.7 (30.2)	29.1 (23.2)	-17.8 (31.6)
	Median	0.38	-0.25	20.0	-20.0	22.5	-15.0
	[Q1, Q3]	[0.00, 1.13]	[-0.63, 0.00]	[10.0, 40.0]	[-40.0, 0.0]	[10.0, 49.0]	[-40.0, 0.0]
	[Min, Max]	[0.00, 2.75]	[-2.25, 1.25]	[0, 90]	[-90, 70]	[0, 99]	[-99, 72]
12 months	n	207	186	199	186	211	204
	Mean (SD)	0.54 (0.65)	-0.33 (0.53)	25.2 (21.4)	-19.4 (27.0)	27.7 (22.3)	-18.3 (29.4)
	Median	0.25	-0.25	20.0	-16.0	20.0	-14.5
	[Q1, Q3]	[0.00, 0.88]	[-0.50, 0.00]	[10.0, 39.0]	[-40.0, 0.0]	[10.0, 40.0]	[-40.0, 0.0]

[Min, Max]	[0.00, 2.88]	[-2.25, 0.88]	[0, 94]	[-100, 57]	[0, 98]	[-98, 65]
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HAQ-DI, Health Assessment Questionnaire-Disability Index, N, total population size; n, sample size; SD, standard deviation; VAS, visual analogue scale

Figure legends

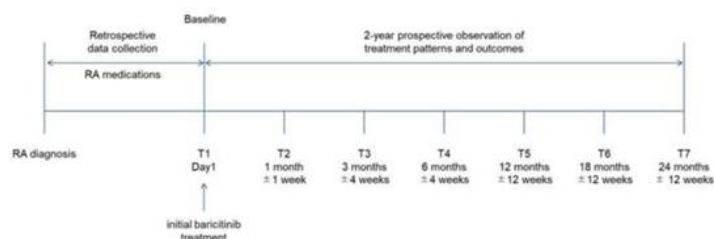


Figure 1. Overall study design

RA, rheumatoid arthritis

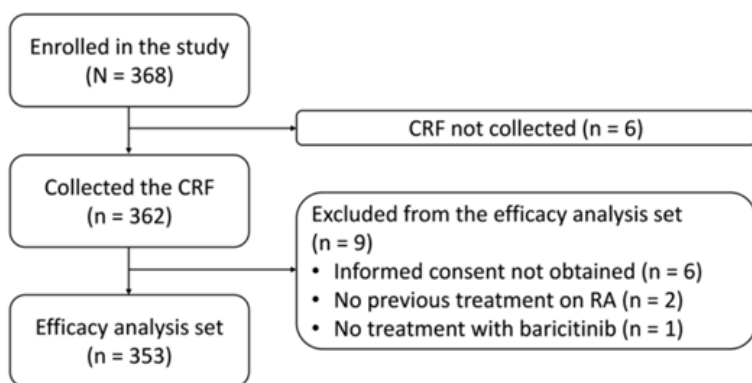


Figure 2. Patient flow diagram

CRF, case report form; N, total population size; n, sample size; RA, rheumatoid arthritis

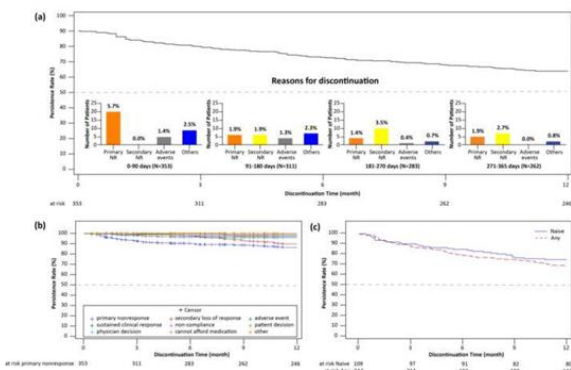


Figure 3. Time to discontinuation

Figure 3A – Overall time to discontinuation, figure 3B – by reasons of discontinuation, figure 3C – by prior b/ts DMARD experience.

Calculated as (discontinuation date - the first dose of baricitinib treatment)/30. Sustained clinical response was not considered as a cause of treatment discontinuation. Discontinuation due to 'other' reasons were considered as 'censored' in figure 3B.

b/tsDMARDs; biological/ targeted synthetic disease-modifying antirheumatic drugs; NR, nonresponse

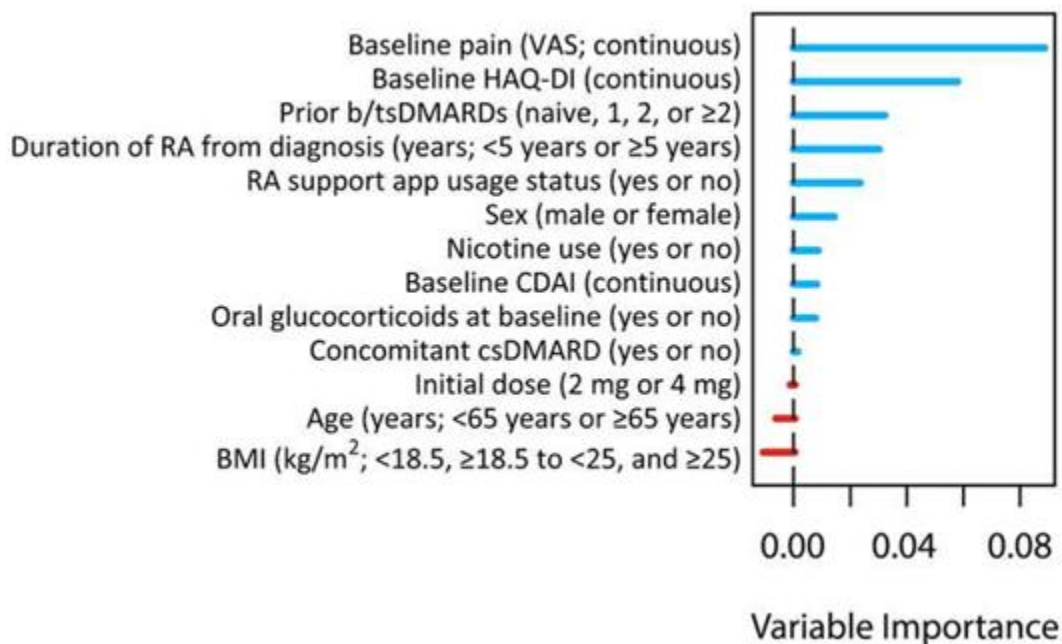


Figure 4. Random forest analysis showing association between patient characteristics and retention

Blue bars indicate the variables that led to better model prediction and red bars indicate the variables that did not.

The variable importance analysis showed that these ten variables are in favour of the model explanation leading to

better model prediction: baseline pain (VAS), baseline HAQ-DI, prior b/tsDMARDs, duration of RA from diagnosis

(years), RA support app usage status, sex, nicotine use, baseline CDAI, oral glucocorticoids at baseline, and

concomitant csDMARD. Patients who discontinued treatment due to sustained clinical response were censored.

BMI, body mass index; CDAI, Clinical Disease Activity Index; csDMARD, conventional synthetic disease-modifying

antirheumatic drugs; Health Assessment Questionnaire-Disability Index, HAQ-DI; RA, rheumatoid arthritis;

b/tsDMARDs; biological/ targeted synthetic disease-modifying antirheumatic drugs; VAS, visual analogue scale